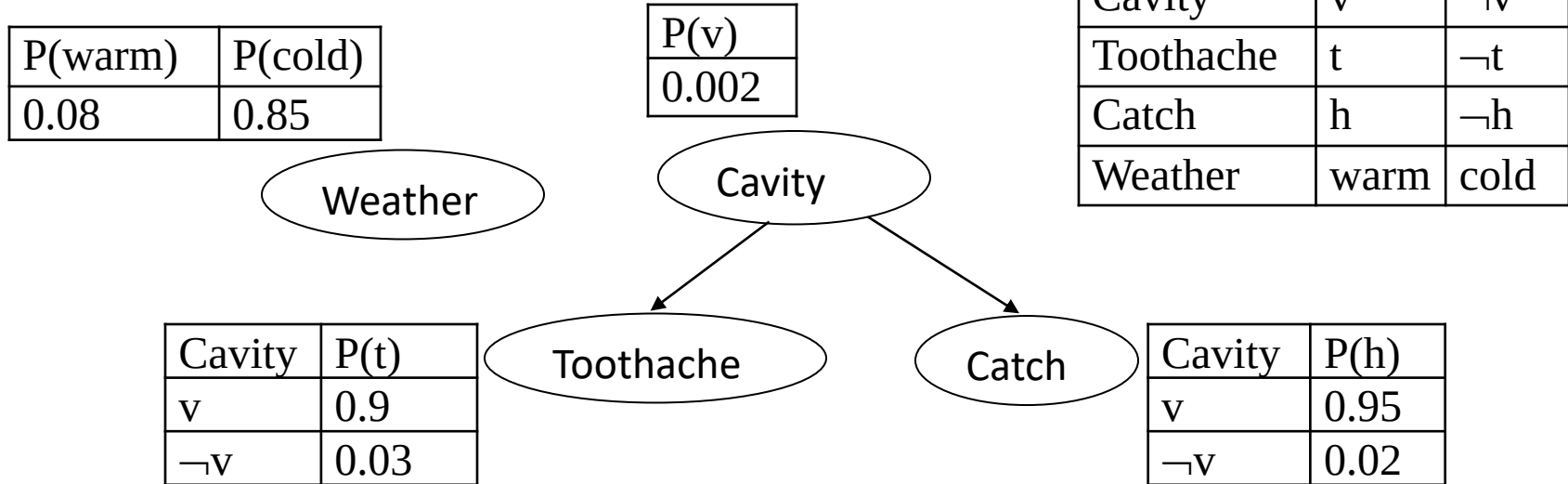


## 6.2. Probabilistic Reasoning using Bayesian Networks

### A. An example of Bayesian Networks



- $P(t \wedge v) = P(t \wedge v \wedge h) + P(t \wedge v \wedge \neg h) = P(t \mid v) * P(v) * P(h \mid v) + \dots = P(t \mid v) * P(v) = 0.9 * 0.002 = \dots$  [No Catch! No Weather, obviously!]
- $P(t \wedge \neg v) = P(t \mid \neg v) * P(\neg v) = 0.03 * 0.998 = \dots$
- $P(\neg h \wedge v) = P(\neg h \mid v) * P(v) = 0.05 * 0.002 = \dots$
- $P(t) = P(t \wedge v \wedge h) + P(t \wedge v \wedge \neg h) + P(t \wedge \neg v \wedge h) + P(t \wedge \neg v \wedge \neg h) = \dots = P(t \mid v) * P(v) + P(t \mid \neg v) * P(\neg v) = \dots$

➤ Formula:  $P((X_1 = x_1) \wedge (X_2 = x_2) \wedge \dots \wedge (X_n = x_n)) =$   
 $P(x_1, x_2, \dots, x_n) = \prod_{i=1:n} P(x_i \mid \text{parents}(X_i))$ , where  $\text{parents}(X_i)$  means  
 specific values of  $\text{Parents}(X_i)$

•  $P(t \vee v) = P(t) + P(v) = \text{-----}$  **[Avoid duplicate atomic events.]**

✓  $P(t \mid (v \vee h)) = ??$

✓  $P(h) = P(t \wedge v \wedge h) + P(\neg t \wedge v \wedge h) + P(t \wedge \neg v \wedge h) + P(\neg t \wedge \neg v \wedge h) = ??$

## B. Specification of a Bayesian Network

1. A BN is a data structure represented by a directed acyclic graph.
2. A node represents a random variable.
3. An arrow (arc) represents a 'parent-child' relationship.
4. A node  $X_i$  is assigned a conditional probability table (CPT) that quantifies the effect of the parents on the node, that is, the distribution,  
$$P(X_i \mid \text{Parents}(X_i)).$$